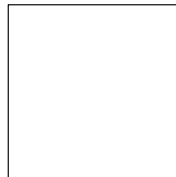
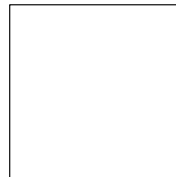


Power



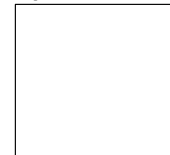
File: power.kicad_sch

MCU



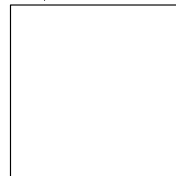
File: mcu.kicad_sch

Lights



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Peripherals



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Battery charger

The schematic diagram illustrates the battery charger circuit for the MAX7792T IC (IC1). The circuit includes the following components and connections:

- IC1 (MAX7792T):** The central integrated circuit with pins labeled B1, B2, F3, F4, F5, F6, A3, C3, B4, C6, B6, A1, A2, B3, E3, A4, A6, E2, E3, B5, A5, and VDD.
- Input and Status:**
 - VBUS is connected to B1 and B2.
 - TP1 (TP_CHG_IN) and TP2 (TP_CHG_STAT) are connected to the CHG_STAT and CHG_INOK pins (A3, C3).
 - TP1 and TP2 are also connected to the INT_CHG, SDA, and SCL pins (B4, C6, B6).
- Resistor Network:**
 - R1 (4k42 1%) and R2 (4k42 1%) are connected to the VCHG and ICHG pins (A1, A2).
 - R3 (10k) is connected to the THM pin (A4) and the +3V3 supply.
 - R7 (10k) is connected to the THM pin (A4) and ground.
- Charge Control:**
 - CHGEN (E3) is connected to the CHGEN pin.
 - CHGEN is also connected to the THM pin (A4) and the +3V3 supply.
- Output and Regulation:**
 - BYP_2 (C2) and BYP_1 (C1) are connected to the BYP pins.
 - BST (F1) is connected to the BST pin.
 - LX_1 (D1), LX_2 (D2), and LX_3 (D3) are connected to the LX pins.
 - SYS_1 (E4), SYS_2 (E5), and SYS_3 (E6) are connected to the SYS pins.
 - PVL (F2) is connected to the PVL pin.
 - BATT_1 (D4), BATT_2 (D5), and BATT_3 (D6) are connected to the BATT pins.
 - CSP (C5) and CSN (C4) are connected to the CSP and CSN pins.
 - BATTSP_/_CSN (C4) is connected to the BATTSP_/_CSN pin.
 - BATTSSN (B5) is connected to the BATTSSN pin.
 - VDD (A5) is connected to the VDD pin.
- Capacitors and Inductors:**
 - C1 (2u2) and C2 (2u2) are connected to the BYP pins.
 - C3 (22u) is connected to the BYP_2 pin.
 - C4 (0u1) is connected to the BST pin.
 - L1 (0u47) is connected to the LX pins.
 - C5 (10u) and C6 (2u2) are connected to the CSP and CSN pins.
 - C7 (2u2) is connected to the BATT pins.
 - C8 (10u) and C9 (10u) are connected to the SYS pins.
 - C10 (2u2) is connected to the VDD pin.
- Other Components:**
 - F1 (MF-USML700/6-2) is the main fuse.
 - BT1 and BT2 are the battery terminals.
 - TH1 (10k) is a thermistor connected to the THM pin (A4) and ground.

Resistor values correspond to 1000mA charge current and 4.47V of charge voltage with empty battery voltage being 3.3V.

When this jumper (JP1) is bridged, multiplexer will always connect data lines to MCU.

USB Multiplexer

The diagram illustrates the circuit for a USB Multiplexer. The central component is the NX3DV221GM_115 IC, labeled U3. Its VCC pin (10) is connected to a +3V3 supply through a 100nF capacitor (C35). The D- and D+ pins (7 and 8) are connected to the D- and D+ pins of a DATAMUX_SEL component. The SDA pin (9) is connected to the DATAMUX_SEL component. The *OE pin (6) is connected to a DATA_OPEN signal through a 10k resistor (R18). The IC's GND pin (5) is connected to ground. The IC has four data outputs: D_MCU- (pin 2), D_MCU+ (pin 1), D_CHG- (pin 4), and D_CHG+ (pin 3).

3V3 rail

VSYS

U4
NCP186AMX330TAG

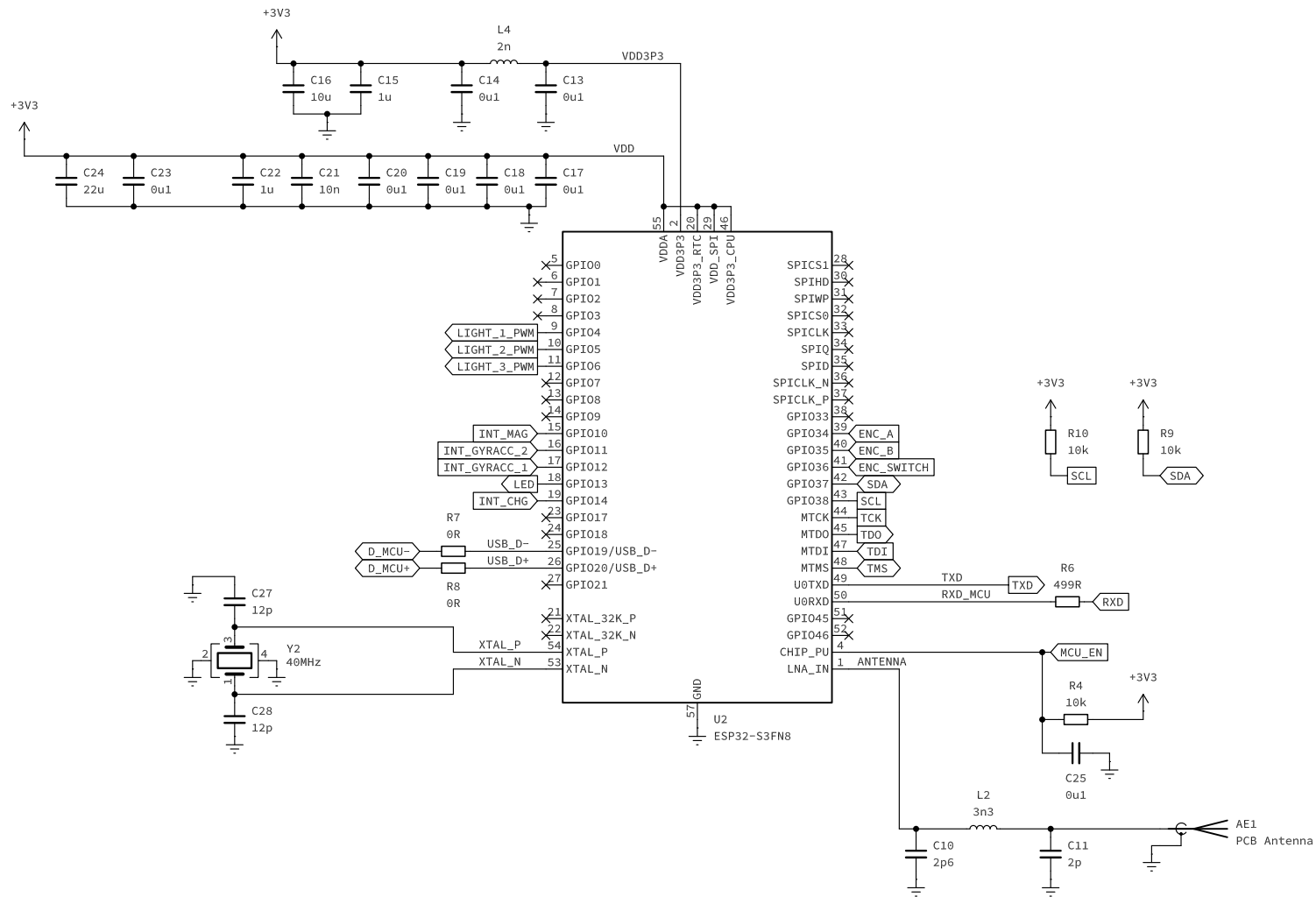
8 IN
7 IN
6 EN
2 OUT
4 FB/ADJ
5 GND
9 EPAD
3 N/C

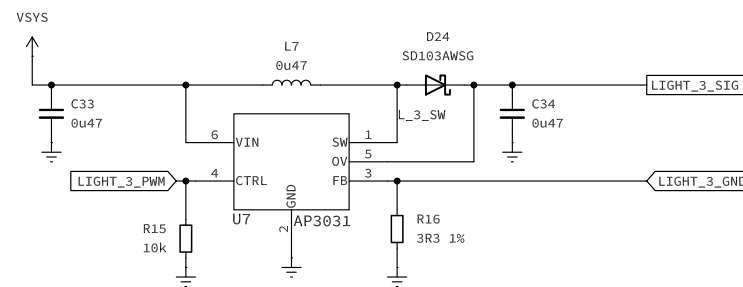
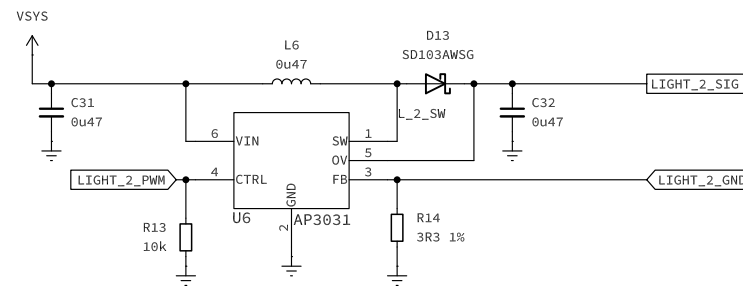
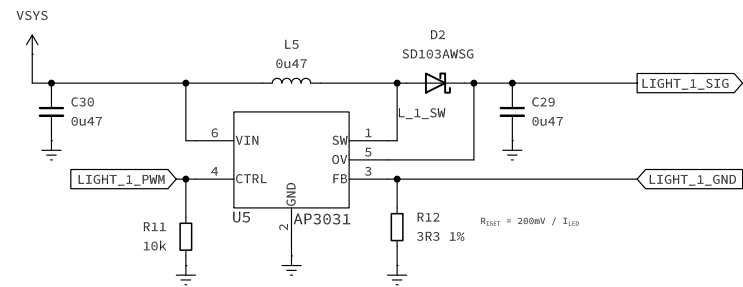
C36
1u

C37
1u

+3V3

[illegible]





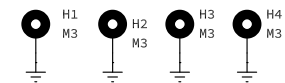


Diagram showing the JTAG pins (TMS, TDI, TDO, TCK) connected to a 4-pin connector labeled J3 and JTAG.

Diagram of the J5 Lights connector showing pin assignments:

- Pin 1: LIGHT_1_SIG
- Pin 2: LIGHT_1_GND
- Pin 3: LIGHT_2_SIG
- Pin 4: LIGHT_2_GND
- Pin 5: LIGHT_3_SIG
- Pin 6: LIGHT_3_GND